EXPERIMENT 14

Cracking the Password Using Hydra

Name: Shaik Reehan	**Reg. No.:** 192565071
Subject: Ethical Hacking	**Course Code:** ITA1413

AIM:

To execute and demonstrate password cracking against an authentication service using THC-Hydra in Kali Linux Operating System, evaluate password vulnerability, and study defensive countermeasures.

1. Hydra (Network Login Cracker):

• *Description:* Hydra is a fast and flexible network logon cracker that supports numerous protocols including SSH, FTP, Telnet, HTTP, MySQL, and RDP. It performs dictionary-based password guessing attacks against remote authentication services to identify weak passwords.
• *Syntax:* hydra [options] -l <username> -P <wordlist> <target_ip> <service>
• *Common Options:*
  • -l <user> : Specifies a single target login username.
  • -L <file> : Specifies a text file containing multiple usernames.
  • -P <file> : Specifies a wordlist file containing passwords to test.
  • -t <num> : Sets the number of parallel tasks/connections (threads) to run (e.g., 4 or 16).
  • -V : Enables verbose output to show attempted login credentials.

2. Defensive Countermeasures Against Password Cracking:

• *Account Lockout:* Lock user accounts temporarily after 3 to 5 failed login attempts to stop automated dictionary attacks.
• *Strong Password Policy:* Mandate passwords with at least 12 characters combining uppercase, lowercase, numbers, and symbols.
• *Multi-Factor Authentication (MFA):* Require a second authentication factor (TOTP / hardware token) to protect accounts if passwords are leaked.
• *Fail2ban / IP Banning:* Automatically block IP addresses exhibiting repeated authentication failures.
• *Disable Password Auth:* Switch SSH from password authentication to public key-based authentication.

OUTPUT:

RESULT:

Thus, password cracking using Hydra was successfully executed in Kali Linux Operating System, verifying password strength testing, dictionary attack behavior, and defensive countermeasures.